Supplementary material for the work "Electron-Ion Path Integral Monte Carlo"

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Supplementary material to the article "Electron-Ion Path Integral Monte Carlo".

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I. INTRODUCTION

We here report the FORTRAN code listing of the code used in the simulations developed in the publication "Electron-Ion Path Integral Monte Carlo". This is shown in Appendix A. The model studied in that publication was that of a 1:1 electron-proton plasma. A particular Two Component Plasma (TCP) where the two species are both fermions, one about 2000 times more heavy than the other. A particular neutral binary mixture with Coulomb pair interactions.

The Restricted Path Integral Monte Carlo (RPIMC) method used in our computer experiments is the one described in Ref. [1, 2]. But the permutation sampling necessary in calculating the determinant required by the study of the Fermi-Dirac statistics was carried out proposing swaps of a randomly chosen pair of particles through the Lévy construction of a pair of Brownian bridges (see Ref. [3] or Section V.G of Ref. [1] and references therein) connecting the two paths.

In the simulation we measure some thermodynamic properties like the total kinetic and potential energy and the superfluid fraction of each species. And some structural properties like the partial Radial Distribution Functions (RDF) for the two species.

The plane waves [1] Path Integral Monte Carlo (PIMC) method used in our computer experiments is the one described in Ref. [1]. A single MC step is made of a swap of a randomly chosen pair of particles through the *Lévy construction* of two Brownian bridge between two randomly chosen timeslices on the two different paths (see Section V.G of Ref. [1] and references therein for the multislice move) and of a displacement of all the timeslices of a particle path chosen at random (the singleslice move). This allows the permutation sampling necessary in calculating the symmetric density matrix required by the study of the Bose-Einstein statistics. We then introduced a path restriction in order to overcome the "fermions sign problem" [5] based on the sign of the free fermions density matrix calculated in subroutine `ifdm`. This results in an approximate RPIMC calculation [2] which calculates the antisymmetric density matrix required by the study of the Fermi-Dirac statistics. Of course we only apply the swap to like particles of the same species.

The extent of the singleslice displacement is chosen at the beginning of the run through the input variable `ALPHA` or it can be adjusted during the run every `IRATIO` MC steps so to have acceptance ratios closest to 1/2 (this functionality is commented out in the present listing).

We discretize the imaginary time between 0 and `BETA` into `FTNO` timeslices of length `LS=BETA/FTNO`. One usually wants to compare simulations at fixed `LS`. The multislice move requires the reconstruction of a randomly chosen number `MBM≤MBMM≤FNTO` of timeslices between a randomly chosen initial timeslice `CP` on a randomly chosen particle `IP` and the final timeslice on a randomly chosen particle `KP`. Here `MBMM` is the maximum number of timeslices in a Brownian bridge. The singleslice move requires the displacement of all the `FTNO` timeslices of the path of the particle `IP` chosen at random. So a single MC step requires a maximum of `2MBMM+FTNO` timeslice moves. In the code we wrote the FORTRAN instructions for the multislice move in lowercase and the ones for the singleslice move in uppercase. The simulation uses only the multislice move if the logical variable `IFBRIDGE` is `TRUE` and only the singleslice move if the logical variable `IFDISP` is `TRUE`. The simulation lasts for a total number of MC steps `NSTEP`. All these variables can be chosen in the `data-mix.in` simulation configuration input file read at startup.

The code works in any space dimension `DIM≤MDIM` with a mixture with a number of components `NMIX≤MMIX`. The number of particles of species `I` is `NPM(I)` and they have mass `FTM(I)`. One has to specify also the thermodynamic configuration giving the density `DENS` and the absolute temperature `TEMP`.

The pair potential between the particles `POTK` is encoded in `SUBROUTINE POT` at the end of the code listing. It includes the AQHS effective potential (`QHS`) and the canonical HS (`BUMP`). Other test cases included are the harmonic potenial (`HARMONIC`), the Coulomb potential (`COULOMBHS` for the hard core Eq. (2.8) in the paper and `COULOMB` without the hard cores but just a cutoff at short distances), the penetrable square well potential (`PSW`), and the Lennard-Jones potential (`LJ`). The ideal gas is obtained choosing the string `FREE` in the `data-mix.in` simulation configuration input file. The potential can be further specified through the variables `RCUT`, `SIG`, `EPSR`, `EPSA`, and `EPS` in the simulation configuration input file.

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[1] For a coherent states PIMC see Ref. [4].

For the sake of the present work we only use the Coulomb potential `COULOMBHS` where `SIG(I,J)` is given by Eq. (2.9) in the main text of the article and `EPSA(I,J)` is given by Eq. (2.10) in the main text of the article.

The simulation calculates both thermodynamic quantities and static structural properties. For the thermodynamic quantities we measure: the kinetic energy `KE`, the potential energy `V`, and the superfluid fraction of each species `SF(I)`. These quantities are stored in the output file `fort.9` every `IPRINT` MC steps. Whereas the path configurations output file `CNFILE` is stored in a *.xyz* format every `ISAVE` MC steps. The first `IEQUI` MC steps are used for the equilibration of the Markov chain. Quantities are only calculated after these equilibration steps. In order to determine efficiently the superfluid fraction at low temperature, which requires large `FTN0`, it is convenient to deactivate the displace move choosing `TRUE` for `IFBRIDGE`. For the structural properties we measure the partial radial distribution function $g_{ij}(r)$ which are written in the output files `rdf11.dat`, `rdf12.dat`, `rdf22.dat`. In the current implementation of the code, the $g_{ij}(r)$ are only calculated during the displacement move every `ICALCG` timeslices.

The logical variable `IFB` is set in the simulation configuration input file. If it is set to `TRUE` permutations of particles are sampled, otherwise only Boltzmann statistics for distinguishable particles is used.

If the variable `INIT` is set to 0 it starts from a path with the particles randomly distributed uniformly with no overlaps with all timeslices copied above the first one for each particle, if it is set to 1 it reads the paths configuration from the file `CNFILE` stored on disk, if it is set to 2 it does a complete restart from the file `CNFILE` and the frozen configuration `fort.20` both stored on disk every `ISAVE` MC steps.

The variable `SEED` is also set in the simulation configuration input file `data-mix.in`. It is the positive integer seed of the pseudo random number generator.

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Appendix A: The code

This is the code used for the PIMC computer experiment. We list here the main FORTRAN code `rpimc-mix.f` with its included `mc-mix.par` parameters file. And the input data file `data-mix.in` (for cases G and H of the Table in the main text of the article) that is read at the beginning of the run.

```
*****************************************************************************
***  rpimc-mix.f  *********************************************************
*****************************************************************************
      PROGRAM RPIMC
      IMPLICIT NONE
C     ****************************************************************
C     ** MONTE CARLO SIMULATION PROGRAM FOR [PW] RESTRICTED PATH      **
C     ** INTEGRAL FOR A BINARY MIXTURE                                **
C     **                                                              **
C     ** space dimensions:    DIM=1,2,3                               **
C     ** number of particles: NP                                      **
C     ** number of timesteps: FTN0                                    **
C     ** units:               hbar=k_B=1                              **
C     ** BETA=1/TEMP=FTN0*LS  (LS imaginary time spacing)             **
C     **                                                              **
C     ** OBSERVABLES:                                                 **
C     ** kinetic energy       KE                                      **
C     ** potential energy     V                                       **
C     ** forces virial        W                                       **
C     ** superfluid fraction  SF                                      **
C     ** radial distribution function in                              **
C     ** 'rdf11.dat', 'rdf12.dat', 'rdf22.dat'                        **
C     ****************************************************************
      INCLUDE       'mc-mix.par'                        ! parameters

      INTEGER*4     SEED
      INTEGER*8     IDIM, STEP, I, J, K
      INTEGER*8     C1, CP, IP, KP, NIP, NKP, LL, OUT1, OUT2, PIP
      INTEGER*8     INIT, NSTEP, NS, IPRINT, ISAVE, IRATIO, IEQUI
      INTEGER*8     MBMM, MCM, PREV(MNP)
      INTEGER*8     RS

      REAL*8        DENS, TEMP, BETA, DENSLJ
      REAL*8        DRMAX, ALPHA, CDIM
      REAL*8        RANF, DUMMY, SR9, SR3
      REAL*8        VLRC, VLRC6, VLRC12, WLRC, WLRC6, WLRC12
      REAL*8        ACM, ACATMA, ACATMAS, ACATMAB
      REAL*8        V, VNEW, VOLD, VEND, DELTV, VN, VS
      REAL*8        W, WNEW, WOLD, WEND, DELTW, WN, WS
      REAL*8        KE, KENEW, KEOLD, DELTKE, KEEND
      REAL*8        AVV, ACV, ACVSQ, FLV
      REAL*8        AVW, ACW, ACWSQ, FLW
      REAL*8        AVKE, ACKE, ACKESQ, FLKE
      REAL*8        ACT, ACTNEW, ACTOLD, DELTACT, DELTACTB
      REAL*8        RXIOLD(MDIM), RXINEW(MDIM)
      REAL*8        RXP(MDIM,0:N), RXPP(MDIM,0:N)
      REAL*8        STD(MMIX), PS, KKK, VVV, WWW
      REAL*8        RATIO, RATIOS, RATIOB
      REAL*8        WIND(MMIX,MDIM), ACWINDSQ(MMIX,MDIM)
      REAL*8        SFI(MMIX), SF(MMIX), WSP(MMIX)
      REAL*8        SGN1, SGN2, RXOLD(MDIM,0:N,MNP)

      INTEGER*8     BIN, MAXBIN, MAXBIN2, ICALCG
      INTEGER*8     HIST11(MBIN),HIST12(MBIN),HIST22(MBIN)
      REAL*8        CONST, RLOWER, RUPPER, TSTEP
      REAL*8        NIDEAL11, NIDEAL12, NIDEAL22
      REAL*8        GR11, GR12, GR22, CO(MMIX)
      REAL*8        CONST11,CONST12,CONST21,CONST22

      CHARACTER     CNFILE*30, POTK*10

      LOGICAL       OVRLAP, IFB, IFDISP, IFBRIDGE

C      PI = ACOS(-1.d0)
      OPEN (UNIT = 11, FILE = 'rdf11.dat', STATUS ='UNKNOWN')
      OPEN (UNIT = 12, FILE = 'rdf12.dat', STATUS ='UNKNOWN')
      OPEN (UNIT = 13, FILE = 'rdf22.dat', STATUS ='UNKNOWN')
      OPEN (UNIT = 14, FILE = 'rdfto.dat', STATUS ='UNKNOWN')
C     ****************************************************************
C     ** READ INPUT DATA                                              **
C     ****************************************************************

      WRITE(*,'('' ********** PROGRAM PIMC ************      ''/)')
      WRITE(*,'('' PLANE WAVES                                  '')')
      WRITE(*,'('' PATH INTEGRAL MONTE CARLO PROGRAM            '')')
      OPEN (UNIT=10, FILE='data-mix.in', STATUS='UNKNOWN')
      READ (10,*) I
      READ (10,*) DIM, NMIX
      READ (10,*) I
      READ (10,*) SEED
      READ (10,*) I
      READ (10,*) IFB
      READ (10,*) I
      READ (10,*) (NPM(J),J=1,NMIX)
      READ (10,*) I
      READ (10,'(A)') POTK
      READ (10,*) I
      READ (10,*) FTN0
      READ (10,*) I
      READ (10,*) (FTM(J),J=1,NMIX)
      READ (10,*) I
      READ (10,*) NSTEP
      READ (10,*) I
      READ (10,*) IPRINT
      READ (10,*) I
      READ (10,*) ISAVE
      READ (10,*) I
      READ (10,*) IEQUI
      READ (10,*) I
      READ (10,*) IRATIO
      READ (10,*) I
      READ (10,*) ICALCG
      READ (10,*) I
      READ (10,'(A)') CNFILE
      READ (10,*) I
      READ (10,*) INIT
      READ (10,*) I
      READ (10,*) DENS
      READ (10,*) I
      READ (10,*) TEMP
      READ (10,*) I
      READ (10,*) ALPHA
      READ (10,*) I
      READ (10,*) MBMM
```

```
      READ (10,*) I
      READ (10,*) RCUT
      READ (10,*) I
      READ (10,*) ((SIG(I, J),I=1,NMIX),J=1,NMIX)
      READ (10,*) I
      READ (10,*) EPSR
      READ (10,*) I
      READ (10,*) ((EPSA(I, J),I=1,NMIX),J=1,NMIX)
      READ (10,*) I
      READ (10,*) EPS
      READ (10,*) I
      READ (10,*) IFDISP
      READ (10,*) I
      READ (10,*) IFBRIDGE
      CLOSE (UNIT=10)

      WRITE(*,'('' IN DIMENSION (1,2,3,...)''I12/)') DIM
      WRITE(*,'('' ************************************      ''/)')

      GOTO 1111                    ! comment if input from keyboard

      WRITE(*,'('' ********** PROGRAM PIMC ************      ''/)')
      WRITE(*,'('' PLANE WAVES                                 ''')')
      WRITE(*,'('' PATH INTEGRAL MONTE CARLO PROGRAM           ''')')

      WRITE(*,'('' ENTER THE SPATIAL AND MIXTURE DIMENSIONS   ''')')
      READ (*,*) DIM, NMIX
      WRITE(*,'('' ENTER SEED FOR RANDOM SEQUENCE              ''')')
      READ (*,*) SEED
      WRITE(*,'('' IF BOSE (T/F)                               ''')')
      READ (*,*) IFB
      WRITE(*,'('' ENTER THE NUMBER OF PARTICLES < MNP         ''')')
      READ (*,*) (NPM(J),J=1,NMIX)
      WRITE(*,'('' ENTER TYPE OF PAIR POTENTIAL (Many Body)   ''')')
      READ (*,'(A)') POTK
      WRITE(*,'('' ENTER THE NUMBER OF DISCRETIZATIONS < N    ''')')
      READ (*,*) FTN0
      WRITE(*,'('' ENTER THE BARE MASS                         ''')')
      READ (*,*) (FTM(J),J=1,NMIX)
      WRITE(*,'('' ENTER NUMBER OF CYCLES                      ''')')
      READ (*,*) NSTEP
      WRITE(*,'('' ENTER NUMBER OF STEPS BETWEEN OUTPUT LINES ''')')
      READ (*,*) IPRINT
      WRITE(*,'('' ENTER NUMBER OF STEPS BETWEEN DATA SAVES   ''')')
      READ (*,*) ISAVE
      WRITE(*,'('' ENTER NUMBER OF STEPS FOR EQUILIBRATION    ''')')
      READ (*,*) IEQUI
      WRITE(*,'('' ENTER INTERVAL FOR UPDATE OF MAX. DISPL.   ''')')
      READ (*,*) IRATIO
      WRITE(*,'('' ENTER INTERVAL FOR RDF CALCULATION          ''')')
      READ (*,*) ICALCG
      WRITE(*,'('' ENTER THE CONFIGURATION FILE NAME           ''')')
      READ (*,'(A)') CNFILE
      WRITE(*,'('' ENTER 0 IF INITIALIZATION NEEDED            ''')')
      READ (*,*) INIT
      WRITE(*,'('' ENTER THE DENSITY                           ''')')
      READ (*,*) DENS
      WRITE(*,'('' ENTER THE TEMPERATURE                       ''')')
      READ (*,*) TEMP
      WRITE(*,'('' ENTER MAX. DISPLACEMENT DRMAX/SIGMA         ''')')
      READ (*,*) ALPHA
      WRITE(*,'('' ENTER MAXIMUM NUMBER OF BRIDGE TIMESLICES  ''')')
      READ (*,*) MBMM
      WRITE(*,'('' ENTER THE POTENTIAL CUTOFF DISTANCE (~ 2.5)''')')
      READ (*,*) RCUT
      WRITE(*,'('' ENTER SIG                                   ''')')
      READ (*,*) ((SIG(I, J),I=1,NMIX),J=1,NMIX)
      WRITE(*,'('' ENTER EPSR                                  ''')')
      READ (*,*) EPSR
      WRITE(*,'('' ENTER EPSA                                  ''')')
      READ (*,*) ((EPSA(I, J),I=1,NMIX),J=1,NMIX)
      WRITE(*,'('' ENTER EPS                                   ''')')
      READ (*,*) EPS
      WRITE(*,'('' IF DISPLACEMENT MOVE (T/F)                  ''')')
      READ (*,*) IFDISP
      WRITE(*,'('' IF BRIDGE MOVE (T/F)                        ''')')
      READ (*,*) IFBRIDGE
      WRITE(*,'('' IN DIMENSION (1,2,3,...)''I12/)') DIM
      WRITE(*,'('' ************************************      ''/)')

 1111 CONTINUE

      IF (MBMM .GT. FTN0) THEN
         WRITE(*,*) "number of timeslices in bridge > ",FTN0
         STOP
      ENDIF

      NP =  0
      DO I=1, NMIX
         NP = NP + NPM(I)
      ENDDO
      DO I=1, NMIX
         CO(I)=DBLE(NPM(I))/DBLE(NP)
      ENDDO

C     ** INITIALIZE RANDOM NUMBER GENERATOR **

      CALL SRAND ( SEED )
      IF (SEED.EQ.1) THEN
         CALL SRAND ( 0 )
      ENDIF

C     ** INVERSE TEMPERATURE BETA **

      BETA = (1.d0/TEMP)

C     ** IMAGINARY TIMESTEP **

      LS  = BETA/FTN0      ! time step
      DO I=1,NMIX
         STD(I) = (LS/FTM(I))**0.5  ! standatd deviation of free-particle rho
      ENDDO

C    ** WRITE INPUT DATA **

      WRITE(*,'('' SEED                      '',I10   )'') SEED
      WRITE(*,'('' POTENTIAL                 '',A     )'') POTK
      WRITE(*,'('' DIMENSIONS                '',I10   )'') DIM
      WRITE(*,'('' NUMBER OF PARTICLES       '',I10   )'') NP
      WRITE(*,'('' NUMBER OF CYCLES          '',I10   )'') NSTEP
      WRITE(*,'('' NUMBER OF EQUIL. STEPS    '',I10   )'') IEQUI
      WRITE(*,'('' OUTPUT FREQUENCY          '',I10   )'') IPRINT
      WRITE(*,'('' SAVE FREQUENCY            '',I10   )'') ISAVE
      WRITE(*,'('' RATIO UPDATE FREQUENCY    '',I10   )'') IRATIO
      WRITE(*,'('' CONFIGURATION FILE  NAME  '',A     )'') CNFILE
      WRITE(*,'('' TEMPERATURE               '',E10.4 )'') TEMP
      WRITE(*,'('' DENSITY                   '',E10.4 )'') DENS
      WRITE(*,'('' PARTICLE MASS             '',E10.4 )'') FTM
      WRITE(*,'('' # TIME SLICES             '',I10   )'') FTN0
      WRITE(*,'('' TIME STEP                 '',E10.4 )'') LS

C    ** CONVERT INPUT DATA TO PROGRAM UNITS **

      SIGMA  = ( DBLE ( NP ) / DENS ) ** ( 1.0 / DIM )
      DRMAX  = ALPHA * SIGMA

      IF (POTK .EQ. 'LJ' .AND. NMIX .EQ. 1) THEN
         DENSLJ = DENS * SIG(1,1) ** DBLE( DIM )
         RCUT = SIGMA/2.d0/SIG(1,1)
      ENDIF

      MAXBIN  = INT ( SIGMA/DELR )
      MAXBIN2 = INT ( MAXBIN/2.0 )
      HIST11  = 0.
      HIST12  = 0.
      HIST22  = 0.

      IF (DIM .eq. 3) THEN
         CONST11 = 4.d0 * PI * DENS * CO(1)/ 3.d0
         CONST12 = 4.d0 * PI * DENS * CO(2)/ 3.d0
         CONST21 = 4.d0 * PI * DENS * CO(1)/ 3.d0
         CONST22 = 4.d0 * PI * DENS * CO(2)/ 3.d0
      ENDIF
      IF (DIM .eq. 2) THEN
         CONST11 = PI * DENS * CO(1)
         CONST12 = PI * DENS * CO(2)
         CONST21 = PI * DENS * CO(1)
         CONST22 = PI * DENS * CO(2)
      ENDIF
      IF (DIM .eq. 1) THEN
         CONST11 = 2.d0 * DENS * CO(1)
         CONST12 = 2.d0 * DENS * CO(2)
         CONST21 = 2.d0 * DENS * CO(1)
         CONST22 = 2.d0 * DENS * CO(2)
      ENDIF

C     ** INITIALIZE CONFIGURATION **

      DO I = 1, NP
         NEXT(I) = I
         LEXT(I) = I
         DO J = 1, FTN0
            MEXT(J,I)  = I
         ENDDO
      ENDDO

C     ** ZERO ACCUMULATORS **

      ACV     = 0.0
      ACVSQ   = 0.0
      FLV     = 0.0
      ACW     = 0.0
      ACWSQ   = 0.0
      FLW     = 0.0
      ACKE    = 0.0
      ACKESQ  = 0.0
      FLKE    = 0.0
      ACWINDSQ= 0.0
      ACM     = 0.0
      ACATMA  = 0.0
      ACATMAB = 0.0
      ACATMAS = 0.0

C     ** READ INITIAL CONFIGURATION **
      IF ( INIT .EQ. 0 ) THEN
         PRINT*,"PATHS INITIALIZATION ..............."
         OPEN (UNIT=9, STATUS='UNKNOWN')
         CALL INITCN ( CNFILE ) ! random configuration
         CALL READCN ( CNFILE ) ! random configuration
         NS = NSTEP
      ELSEIF ( INIT .EQ. 1) THEN
         PRINT*,"RESTART FROM CONFIG  ..............."
         OPEN (UNIT=9, STATUS='UNKNOWN')
         CALL READCN ( CNFILE )
         NS = NSTEP
      ELSEIF ( INIT .EQ. 2) THEN
         PRINT*,"RESTART FULL         ..............."
         OPEN (UNIT=9, ACCESS='APPEND')
         CALL READCN ( CNFILE )
         ACATMA  = 0.0
         ACATMAB = 0.0
         ACATMAS = 0.0
         READ(20,*)(HIST11(I),I=1,MAXBIN),
     &              (HIST12(I),I=1,MAXBIN),
     &              (HIST22(I),I=1,MAXBIN),
     &          ACV,ACVSQ,ACW,ACWSQ,ACKE,ACKESQ,ACWINDSQ,ACM,NS
         NS=NS+NSTEP
      ELSE
         WRITE (*,*) 'INIT different from 0,1,2'
         STOP
      ENDIF
```

```
C     ** CALCULATE LONG RANGE CORRECTIONS    **
C     ** SPECIFIC TO THE LENNARD JONES FLUID **
        IF (DIM .EQ. 1) CDIM = 1
        IF (DIM .EQ. 2) CDIM = PI
        IF (DIM .EQ. 3) CDIM = 2*PI

        SR3 = - RCUT ** (- 6. + DIM)/(-6. + DIM)
        SR9 = - RCUT ** (- 12. + DIM)/(-12. + DIM)

        VLRC12 =   4 * EPS * CDIM * DENSLJ * NP * SR9
        VLRC6  = - 4 * EPS * CDIM * DENSLJ * NP * SR3
        VLRC   =   VLRC12 + VLRC6
        WLRC12 =   4.0  * VLRC12
        WLRC6  =   2.0  * VLRC6
        WLRC   =   WLRC12 + WLRC6

C     ** WRITE OUT SOME USEFUL INFORMATION **

        WRITE(*,'('' SIGMA                     '',E10.4)') SIGMA
        WRITE(*,'('' MAXIMUM DISPLACEMENT      '',E10.4)') DRMAX

C    ** CALCULATE INITIAL ENERGY AND CHECK FOR OVERLAPS **

        CALL SUMUP (POTK, OVRLAP, KE, V, W)

        IF (POTK .EQ. 'LJ') THEN
           VS = ( V + VLRC )
           WS = ( W + WLRC )
        ELSE
           VS = V
           WS = W
        ENDIF

        WRITE(*,'('' INITIAL V                 '',E10.4)') VS
        WRITE(*,'('' INITIAL W                 '',E10.4)') WS
        WRITE(*,'('' INITIAL KE                '',E10.4)') KE

        WRITE(*,'(//'' START OF MARKOV CHAIN               ''//)')
        WRITE(*,'('' NMOVE     RATIO       ACTION          ''/)')

C     ****************************************************************
C     ** LOOPS OVER ALL CYCLES AND ALL TIME SLICES                   **
C     ****************************************************************

        DO 100 STEP = NS-NSTEP+1, NS

           CP = INT(FTN0*RANF(DUMMY))+1    ! select a random timeslice
C           RS = INT(FTN0*RANF(DUMMY))+1 ! select a random referenceslice
           RS = 1
           MBM = INT((MBMM-1)*RANF(DUMMY))+2 ! # timeslices in bridge
           IP = INT(NP*RANF(DUMMY))+1      ! select a particle
           KP = INT(NP*RANF(DUMMY))+1      ! select another particle
           IF (.NOT.IFB) KP = IP           ! for boltzmann statistics
           IF (IFDISP) GOTO 77   ! uncomment if only displacement move
           IF (ISP(IP).NE.ISP(KP)) GOTO 77 ! different species
C     ****************************************************************
C     ** BRIDGE&SWAP MOVE (BOSE STATISTICS)                          **
C     ****************************************************************
           mcm = cp+mbm-floor((cp+mbm-.1)/ftn0)*ftn0

           call equiv(rxold,rx)

           nip=next(ip)
           nkp=next(kp)
           ps=0.d0

           vold=0.d0
           wold=0.d0
           if(cp+mbm.le.ftn0)then
              do i=cp+1,mcm-1
                 call pppnergy ( potk, rx(:,i,ip), ip,
     :                rx(:,i,kp), kp, i,
     :                vvv, www )
                 vold=vold+vvv
                 wold=wold+www
              enddo
           else
              do i=cp+1,ftn0
                 call pppnergy ( potk, rx(:,i,ip), ip,
     :                rx(:,i,kp), kp, i,
     :                vvv, www )
                 vold=vold+vvv
                 wold=wold+www
              enddo
              do i=1,mcm-1
                 call pppnergy ( potk, rx(:,i,nip), nip,
     :                rx(:,i,nkp), nkp, i,
     :                vvv, www )
                 vold=vold+vvv
                 wold=wold+www
              enddo
           endif

           keold=0.d0
           do k=1,np
              if(k.eq.ip.or.k.eq.kp.or.k.eq.nip.or.k.eq.nkp)then
                 do i=1,ftn0
                    call kkknergy ( rx(:,i,k), k, i,
     :                   kkk )
                    keold=keold+kkk
                 enddo
              endif
           enddo

           if(cp+mbm.le.ftn0)then
              call bridge(rx(:,cp,ip),rx(:,mcm,kp),std(isp(ip)),
     :            rxp,out1)
              if (ip.ne.kp) then
                 call bridge(rx(:,cp,kp),rx(:,mcm,ip),std(isp(ip)),
     :                rxpp,out2)
              endif
           else
              call bridge(rx(:,cp,ip),rx(:,mcm,nkp),std(isp(ip)),
     :            rxp,out1)
              if (ip.ne.kp) then
                 call bridge(rx(:,cp,kp),rx(:,mcm,nip),std(isp(ip)),
     :                rxpp,out2)
              endif
           endif

           vnew=0.d0
           wnew=0.d0
           ll=0
           if(cp+mbm.le.ftn0)then
              do i=cp+1,mcm-1
                 ll=ll+1
                 if (ip.eq.kp) then
                    call pppnergy ( potk, rxp(:,ll), ip,
     :                   rxp(:,ll), kp, i,
     :                   vvv, www )
                 else
                    call pppnergy ( potk, rxp(:,ll), ip,
     :                   rxpp(:,ll), kp, i,
     :                   vvv, www )
                 endif
                 vnew=vnew+vvv
                 wnew=wnew+www
              enddo
           else
              do i=cp+1,ftn0
                 ll=ll+1
                 if (ip.eq.kp) then
                    call pppnergy ( potk, rxp(:,ll), ip,
     :                   rxp(:,ll), kp, i,
     :                   vvv, www )
                 else
                    call pppnergy ( potk, rxp(:,ll), ip,
     :                   rxpp(:,ll), kp, i,
     :                   vvv, www )
                 endif
                 vnew=vnew+vvv
                 wnew=wnew+www
              enddo
              do i=1,mcm-1
                 ll=ll+1
                 if (ip.eq.kp) then
                    call pppnergy ( potk, rxp(:,ll), nip,
     :                   rxp(:,ll), nkp, i,
     :                   vvv, www )
                 else
                    call pppnergy ( potk, rxpp(:,ll), nip,
     :                   rxp(:,ll), nkp, i,
     :                   vvv, www )
                 endif
                 vnew=vnew+vvv
                 wnew=wnew+www
              enddo
           endif

           ps=ps+vnew-vold

           call acc_p(ps,ip,kp,cp)
           if (ps.gt.ranf(dummy)) then

              acatmab = acatmab + 1.d0
              call update(cp,rxp,ip,nkp)
              if (ip.ne.kp) then
                 call update(cp,rxpp,kp,nip)
                 call swap(cp,ip,nip,kp,nkp)
                 call switch(next(ip),next(kp))
                 call switch(mext(cp,ip),mext(cp,kp))
                 acatmas = acatmas + 1.d0
              endif

c     restricted path integral
              do i=1,ftn0
                 sgn1=1.
                 do j=1,nmix
                    call ifdm(j,rs,i,sgn2)
                    sgn1=sgn1*sgn2
                 enddo
                 if ( sgn1 .lt. 0. ) goto 1234
              enddo

              kenew=0.d0
              do k=1,np
                 if(k.eq.ip.or.k.eq.kp.or.k.eq.nip.or.k.eq.nkp)then
                    do i=1,ftn0
                       call kkknergy ( rx(:,i,k), k, i,
     :                      kkk )
                       kenew=kenew+kkk
                    enddo
                 endif
              enddo
              ke=ke+kenew-keold

              v=v+vnew-vold
              w=w+wnew-wold
           endif
           goto 1235
 1234      continue
           call equiv(rx,rxold)
           acatmab = acatmab - 1.0
           if (ip.ne.kp) then
              acatmas = acatmas - 1.0
              call switch(next(ip),next(kp))
              call switch(mext(cp,ip),mext(cp,kp))
           endif
 1235      continue

c     build the permutations cycles in lext
```

```
      call permcyc()

      IF (STEP.GT.IEQUI) THEN
         ACM = ACM + 1.0

C     ** CALCULATE INSTANTANEOUS VALUES **

            IF (POTK .EQ. 'LJ') THEN
               VN = ( V + VLRC )
            ELSE
               VN = V
               WN = W
            ENDIF
            CALL CWIND( ISP(IP), WIND )

C     ** ACCUMULATE AVERAGES **

         ACV    = ACV    + VN
         ACVSQ  = ACVSQ  + VN*VN
         ACW    = ACW    + WN
         ACWSQ  = ACWSQ  + WN*WN
         ACKE   = ACKE   + KE
         ACKESQ = ACKESQ + KE*KE
         DO J=1,NMIX
            DO I=1,DIM
               ACWINDSQ(J,I) = ACWINDSQ(J,I) + WIND(J,I)**2.
            ENDDO
         ENDDO
      ENDIF

      IF (IFBRIDGE) GOTO 97

C     ****************************************************************
C     ** ENDS BRIDGE&SWAP MOVE                                      **
C     ****************************************************************

 77   CONTINUE

C     ****************************************************************
C     ** DISPLACEMENT MOVE                                          **
C     ****************************************************************

C     ** PREVIOUS IP **

      DO I=1,NP
         J=NEXT(I)
         PREV(J)=I
      ENDDO

      NIP = NEXT(IP)
      PIP = PREV(IP)

      DO 90 C1 = 1, FTNO

         DO IDIM = 1, DIM
            RXIOLD(IDIM) = RX(IDIM,C1,IP)
         ENDDO

C     ** CALCULATE THE ENERGY OF I IN THE OLD CONFIGURATION **

         CALL PENERGY ( POTK, RXIOLD, IP, C1,
     :          VOLD, WOLD )

         CALL KENERGY ( RXIOLD, IP, C1,
     :          KEOLD )

C     ** INSTANTANEOUS VALUE OF THE ACTION **

         ACTOLD = KEOLD + VOLD

C     ** MOVE I AND PICKUP THE CENTRAL IMAGE **

         DO IDIM = 1, DIM
            RXINEW(IDIM) = RXIOLD(IDIM) +
     :            ( 2.0 * RANF ( DUMMY ) - 1.0 )*DRMAX
c            RXINEW(IDIM) = RXINEW(IDIM) -
c     :            DNINT ( RXINEW(IDIM)/SIGMA )*SIGMA
         ENDDO

C     ** CALCULATE THE ENERGY OF I IN THE NEW CONFIGURATION **

         CALL PENERGY ( POTK, RXINEW, IP, C1,
     :          VNEW, WNEW )

         CALL KENERGY ( RXINEW, IP, C1,
     :          KENEW )

C     ** INSTANTANEOUS VALUE OF THE ACTION **

         ACTNEW = KENEW + VNEW

         DELTV  = VNEW  - VOLD
         DELTW  = WNEW  - WOLD
         DELTKE = KENEW - KEOLD

C     ** CHECK FOR ACCEPTANCE **

         DELTACT  = ACTNEW - ACTOLD
         DELTACTB = LS*DELTACT

         IF ( EXP ( - DELTACTB ) .GT. RANF ( DUMMY ) ) THEN
            V          = V  + DELTV
            W          = W  + DELTW
            KE         = KE + DELTKE
            ACATMA = ACATMA + 1.0
            DO IDIM = 1, DIM
               RX(IDIM,C1,IP)  = RXINEW(IDIM)
C     IMAGINARY TIME PERIODIC BOUNDARY CONDITIONS
               IF (C1.EQ.1) THEN
                  RX(IDIM,FTNO+1,PIP)=RX(IDIM,C1,IP)
               ENDIF
               IF (C1.EQ.FTNO) THEN
                  RX(IDIM,0,NIP)=RX(IDIM,C1,IP)
               ENDIF
            ENDDO

C     RESTRICTED PATH INTEGRAL
            DO I=1,FTNO
               SGN1=1.
               DO J=1,NMIX
                  CALL IFDM(J,RS,I,SGN2)
                  SGN1=SGN1*SGN2
               ENDDO
               IF ( SGN1 .LT. 0. ) GOTO 1236
            ENDDO
            GOTO 1237
 1236       CONTINUE
            V          = V  - DELTV
            W          = W  - DELTW
            KE         = KE - DELTKE
            ACATMA = ACATMA - 1.0

            DO IDIM = 1, DIM
               RX(IDIM,C1,IP)  = RXIOLD(IDIM)
               IF (C1.EQ.1) THEN
                  RX(IDIM,FTNO+1,PIP)=RX(IDIM,C1,IP)
               ENDIF
               IF (C1.EQ.FTNO) THEN
                  RX(IDIM,0,NIP)=RX(IDIM,C1,IP)
               ENDIF
            ENDDO
 1237       CONTINUE
         ENDIF

         IF (STEP.GT.IEQUI) THEN
            ACM = ACM + 1.0

C     ** CALCULATE INSTANTANEOUS VALUES **

C     HISTOGRAM FOR g(r)
            IF ( MOD ( C1, ICALCG ) .EQ. 0 ) THEN
               CALL GOFR(C1, DENS, MAXBIN, HIST11, HIST12, HIST22)
            ENDIF

            IF (POTK .EQ. 'LJ') THEN
               VN = ( V + VLRC )
            ELSE
               VN = V
               WN = W
            ENDIF
            CALL CWIND( ISP(IP),WIND )

C     ** ACCUMULATE AVERAGES **

            ACV    = ACV    + VN
            ACVSQ  = ACVSQ  + VN*VN
            ACW    = ACW    + WN
            ACWSQ  = ACWSQ  + WN*WN
            ACKE   = ACKE   + KE
            ACKESQ = ACKESQ + KE*KE
            DO J=1,NMIX
               DO I=1,DIM
                  ACWINDSQ(J,I) = ACWINDSQ(J,I) + WIND(J,I)**2.
               ENDDO
            ENDDO
         ENDIF

C     ****************************************************************
C     ** END DISPLACEMENT MOVE                                      **
C     ****************************************************************

 90   CONTINUE

C     ****************************************************************
C     ** ENDS LOOP OVER TIME SLICES                                 **
C     ****************************************************************

 97   CONTINUE

C     ** WRITE OUT THE INSTANTANEOUS VALUES ON FORT.9 **

      SF  = 0.0
      SFI = 0.0
      DO I=1, NMIX
         DO IDIM = 1, DIM
            SFI(I) = SFI(I) + WIND(I,IDIM)*WIND(I,IDIM)
            SF(I)  = SF(I)  + ACWINDSQ(I,IDIM)
         ENDDO
      ENDDO
      DO I=1, NMIX
         SFI(I) = SFI(I)*FTM(I)/(DIM*BETA*NPM(I))
         SF(I)  = SF(I)*FTM(I)/(DIM*BETA*NPM(I)*ACM)
      ENDDO
      IF (ISP(IP).EQ.1)THEN
         WSP(1)=1
         WSP(2)=0
      ELSE
         WSP(1)=0
         WSP(2)=1
      ENDIF

      CALL FLUSH(9)
      IF ( MOD ( STEP, IPRINT ) .EQ. 0 ) THEN
         WRITE(9,*) STEP,DIM*NP/2/LS-KE/FTNO,V/FTNO,W/FTNO,
     :              SFI,WSP
C         WRITE(9,*) STEP, DIM*NP/2/LS-KE/FTNO, V/FTNO, W/FTNO
      ENDIF

C     ** CREATE POSITION DISTRIBUTION **

C        CALL DISTR(30_8,STEP*FTNO*NP/30)
```

```
C     ** PERFORM PERIODIC OPERATIONS **

          IF ( MOD ( STEP, IRATIO ) .EQ. 0 ) THEN

C     ** ADJUST MAXIMUM DISPLACEMENT **

             RATIO  = ACATMA  / DBLE ( FTN0 * IRATIO )
             RATIOB = ACATMAB / DBLE ( IRATIO )
             RATIOS = ACATMAS / DBLE ( IRATIO )

             IF ( RATIO .GT. 0.5 ) THEN

C                DRMAX  = DRMAX  * 1.05

             ELSE

C                DRMAX  = DRMAX  * 0.95

             ENDIF

             ACATMA  = 0.0
             ACATMAB = 0.0
             ACATMAS = 0.0

          ENDIF

          IF ( MOD ( STEP, IPRINT ) .EQ. 0 ) THEN

C     ** WRITE OUT RUNTIME INFORMATION **

             WRITE(*,*) '          step            acm  dr/sig    ard
     :     arb       ars'
             WRITE(*,'(2(2XI13),(2Xf6.4),4(2Xf8.6))') STEP, INT(ACM),
     :                                 ALPHA, RATIO, RATIOB, RATIOS
             WRITE(*,*) '            ke              v              w
     :        sf'
             WRITE(*,'(4(2XE13.7))') KE/FTN0, V/FTN0, W/FTN0, SFI
             WRITE(*,*) '          <ke>            <v>            <w>
     :         <sf>'
             WRITE(*,'(4(2XE13.7))') ACKE/ACM/FTN0, ACV/ACM/FTN0,
     :                               ACW/ACM/FTN0, SF

          ENDIF

          IF ( MOD ( STEP, ISAVE ) .EQ. 0 ) THEN

C     ** WRITE OUT THE CONFIGURATION AT INTERVALS **

             CALL WRITCN ( CNFILE )
             REWIND(20)
             CALL FLUSH(20)
             WRITE(20,*)(HIST11(I),I=1,MAXBIN),
     &                  (HIST12(I),I=1,MAXBIN),
     &                  (HIST22(I),I=1,MAXBIN),
     &            ACV,ACVSQ,ACW,ACWSQ,ACKE,ACKESQ,ACWINDSQ,ACM,STEP

C     ** WRITE OUT THE g(r) **
             IF ( STEP .GT. IEQUI ) THEN
                TSTEP = DBLE((STEP - IEQUI)*INT(FTN0/ICALCG))
                REWIND(11)
                REWIND(12)
                REWIND(13)
                REWIND(14)
                DO BIN = 1, MAXBIN2
                   RLOWER = DBLE ( BIN - 1 ) * DELR
                   RUPPER = RLOWER + DELR
                   NIDEAL11 = CONST11*( RUPPER**DIM - RLOWER**DIM )
                   NIDEAL12 = CONST12*( RUPPER**DIM - RLOWER**DIM )
                   NIDEAL22 = CONST22*( RUPPER**DIM - RLOWER**DIM )
                   GR11 = DBLE(HIST11(BIN))/TSTEP/DBLE(NPM(1))/NIDEAL11
                   GR12 = DBLE(HIST12(BIN))/TSTEP/DBLE(NPM(1))/NIDEAL12
                   GR22 = DBLE(HIST22(BIN))/TSTEP/DBLE(NPM(2))/NIDEAL22
                   CALL FLUSH(11)
                   CALL FLUSH(12)
                   CALL FLUSH(13)
                   CALL FLUSH(14)
                   WRITE (11,*) DBLE(BIN-1)*DELR, GR11
                   WRITE (12,*) DBLE(BIN-1)*DELR, GR12
                   WRITE (13,*) DBLE(BIN-1)*DELR, GR22
                   WRITE (14,*) DBLE(BIN-1)*DELR,
     :             CO(1)**2*GR11+2.d0*CO(1)*CO(2)*GR12+CO(2)**2*GR22
                ENDDO
             ENDIF

          ENDIF

 100    CONTINUE

C     ****************************************************************
C     ** ENDS THE LOOP OVER CYCLES                                  **
C     ****************************************************************

        WRITE(*,'(//'' END OF MARKOV CHAIN          ''//)')

C     ** CHECKS FINAL VALUE OF THE POTENTIAL ENERGY IS CONSISTENT **

        CALL SUMUP (POTK, OVRLAP, KEEND, VEND, WEND)

        IF ( ABS(VEND - V) .GT. 1.0d-03 ) THEN

           WRITE(*,'(''  PROBLEM WITH V ENERGY !!!'')')
           WRITE(*,'('' VEND  = '', E20.6)') VEND
           WRITE(*,'('' V     = '', E20.6)') V

        ENDIF

        IF ( ABS(KEEND - KE) .GT. 1.0d-03 ) THEN

           WRITE(*,'(''  PROBLEM WITH KE ENERGY !!!'')')
           WRITE(*,'('' KEEND  = '', E20.6)') KEEND
           WRITE(*,'('' KE     = '', E20.6)') KE

        ENDIF

C     ** WRITE OUT THE FINAL CONFIGURATION FROM THE RUN **

        CALL WRITCN ( CNFILE )

C     ** CALCULATE AND WRITE OUT RUNNING AVERAGES **

        AVV   = ACV / ACM
        ACVSQ = ( ACVSQ / ACM ) - AVV ** 2
        AVKE   = ACKE / ACM
        ACKESQ = ( ACKESQ / ACM ) - AVKE ** 2

C    ** CALCULATE FLUCTUATIONS **

        IF ( ACVSQ  .GT. 0.0 ) FLV  = SQRT ( ACVSQ/ACM )/FTN0
        IF ( ACKESQ .GT. 0.0 ) FLKE = SQRT ( ACKESQ/ACM )/FTN0

        WRITE(*,'(/'' AVERAGES ''/ )')
        WRITE(*,'(''  <V/N>                 = '',E12.6)') AVV/FTN0
        WRITE(*,'(''  <KE/N>                = '',E12.6)') AVKE/FTN0

        WRITE(*,'(/'' FLUCTUATIONS ''/)')

        WRITE(*,'(''  FLUCTUATION IN <V/N>  = '',E12.6)') FLV
        WRITE(*,'(''  FLUCTUATION IN <KE/N> = '',E12.6)') FLKE
        WRITE(*,'(/'' END OF SIMULATION ''/)')

        CLOSE (UNIT = 9 )
        CLOSE (UNIT = 11)
        CLOSE (UNIT = 12)
        CLOSE (UNIT = 13)
        CLOSE (UNIT = 14)
C     ****************************************************************
C     ****************************************************************
C     ****************************************************************
C     ** THE END                                                    **
C     ****************************************************************
C     ****************************************************************
C     ****************************************************************
        STOP
        END


        subroutine permcyc()
        implicit none
!     given a particle i lext(i) returns the cycle it belongs to
        INCLUDE       'mc-mix.par'

        integer*8 i,j,k,ii(mnp),jj,kk,ll

        kk=1
        do i=1,np
           ll=1
           ii(ll)=i
 1         jj=next(ii(ll))
           do j=1,ll
              if(jj.eq.ii(j))goto 2
           enddo
           ll=ll+1
           ii(ll)=jj
           goto 1
 2         do k=1,ll
              lext(ii(k))=kk
           enddo
           kk=kk+1
        enddo

        return
        end


        subroutine switch(i,j)
!     switch i and j
        implicit none
        integer*8 i,j,k
        k=i
        i=j
        j=k
        return
        end


        subroutine acc_p(p,ip,kp,j)
        implicit none
!     complete acceptance probability
        INCLUDE       'mc-mix.par'

        integer*8 ip,kp,j,idim
        real*8 p, rho
        real*8 rxink,rxnik
        real*8 rxini,rxknk
        integer*8 mcm

        p=exp(-ls*p)         ! contribution from the pair potential
        if (ip.eq.kp) return

        mcm = j+mbm-floor((j+mbm-.1)/ftn0)*ftn0
        rho=0.d0
        do idim=1,dim
           if(j+mbm.le.ftn0)then
              rxink=rx(idim,j,ip)-rx(idim,mcm,kp)
              rxnik=rx(idim,j,kp)-rx(idim,mcm,ip)
              rxini=rx(idim,j,ip)-rx(idim,mcm,ip)
              rxknk=rx(idim,j,kp)-rx(idim,mcm,kp)
              rxink=rxink-dnint(rxink/sigma)*sigma
              rxnik=rxnik-dnint(rxnik/sigma)*sigma
              rxini=rxini-dnint(rxini/sigma)*sigma
              rxknk=rxknk-dnint(rxknk/sigma)*sigma
           else
```

```
            rxink=rx(idim,j,ip)-rx(idim,mcm,next(kp))
            rxnik=rx(idim,j,kp)-rx(idim,mcm,next(ip))
            rxini=rx(idim,j,ip)-rx(idim,mcm,next(ip))
            rxknk=rx(idim,j,kp)-rx(idim,mcm,next(kp))
            rxink=rxink-dnint(rxink/sigma)*sigma
            rxnik=rxnik-dnint(rxnik/sigma)*sigma
            rxini=rxini-dnint(rxini/sigma)*sigma
            rxknk=rxknk-dnint(rxknk/sigma)*sigma
         endif
         rho=rho+rxink**2+rxnik**2-rxini**2-rxknk**2
      enddo
      rho=ftm(isp(ip))*rho/(2.*mbm*ls)
      p=p*exp(-rho)
      return
      end


      subroutine update(j,rxp,ip,nip)
      implicit none
!     updates a portion of the current path x using the proposed path xp
      INCLUDE       'mc-mix.par'

      integer*8 ip,nip,kp,nkp,j,l,k,idim
      integer*8 mcm
      real*8 rxp(mdim,0:n)

      mcm = j+mbm-floor((j+mbm-.1)/ftn0)*ftn0

      l=0
      if(j+mbm.le.ftn0)then
         do k=j+1,mcm-1
            l=l+1
            do idim=1,dim
               rx(idim,k,ip)=rxp(idim,l)
            enddo
         enddo
      else
         do k=j+1,ftn0
            l=l+1
            do idim=1,dim
               rx(idim,k,ip)=rxp(idim,l)
            enddo
         enddo
         do k=1,mcm-1
            l=l+1
            do idim=1,dim
               rx(idim,k,nip)=rxp(idim,l)
            enddo
         enddo
      endif
      do idim=1,dim
         rx(idim,ftn0+1,ip)=rx(idim,1,nip)
         rx(idim,0,nip)=rx(idim,ftn0,ip)
      enddo

      return
      end


      subroutine swap(j,ip,nip,kp,nkp)
      implicit none
!     updates a portion of the current path x using the proposed path xp
      INCLUDE       'mc-mix.par'

      integer*8 ip,nip,kp,nkp,j,k,idim
      integer*8 mcm
      real*8    rr

      mcm = j+mbm-floor((j+mbm-.1)/ftn0)*ftn0

      if(j+mbm.le.ftn0)then
         do k=mcm,ftn0
            do idim=1,dim
               rr=rx(idim,k,ip)
               rx(idim,k,ip)=rx(idim,k,kp)
               rx(idim,k,kp)=rr
            enddo
         enddo
         do idim=1,dim
            rx(idim,ftn0+1,ip)=rx(idim,1,nkp)
            rx(idim,ftn0+1,kp)=rx(idim,1,nip)
            rx(idim,0,nip)=rx(idim,ftn0,kp)
            rx(idim,0,nkp)=rx(idim,ftn0,ip)
         enddo
      endif
      return
      end


      subroutine bridge(x0,x1,std,xnew,out)
      implicit none
!     sample m gaussians with std from xnew(0)=x0 to xnew(ftn0)=x1
      INCLUDE       'mc-mix.par'

      integer*8 l1,l2,l3,j,out,idim
      real*8 std,d,s,xi
      real*8 x0(mdim),x1(mdim),xnew(mdim,0:n)

      out=0
      l3=mbm
      do idim=1,dim
         xnew(idim,0)=x0(idim)
         xnew(idim,l3)=x0(idim)+((x1(idim)-x0(idim))-
     :          dnint((x1(idim)-x0(idim))/sigma)*sigma)
      enddo
      do j=1,mbm-1
         l1=j-1
         l2=j
         s=std*(dble(l3-l2)/dble(l3-l1))**0.5d0
         do idim=1,dim
            d=xnew(idim,l3)-xnew(idim,l1)
            d=d-dnint(d/sigma)*sigma
            xnew(idim,j)=xnew(idim,l1)+d/dble(l3-l1)+xi(s)
c            if (xnew(idim,j).gt.sigma/2.or.
c     :             xnew(idim,j).lt.-sigma/2) then
c               out=1
c               return
c            endif
c            xnew(idim,j)=xnew(idim,j)-dnint(xnew(idim,j)/sigma)*sigma
         enddo
      enddo
      return
      end


      function xi(std)
!     sample a gaussian with standard deviation std (box-muller method)
      implicit none
      real*8 xi,std,pi,ranf
      data pi/3.14159265358979323846264338328d0/
      xi=cos(pi*ranf(0.d0))*std*sqrt(-2.d0*log(tiny(pi)+ranf(0.d0)))
      return
      end


      SUBROUTINE DISTR(NN,NORM)
      IMPLICIT NONE
C     WRITES ON FORT.10 THE X-POSITION DISTRIBUTION
      INCLUDE       'mc-mix.par'

      REAL*8        DS,DIST(0:1000)
      INTEGER*8     NN,NORM,I,J,K
      SAVE          DIST

      DS=SIGMA/NN

      DO I=0,NN
         DO J=1,FTN0
            DO K=1,NP
               IF(-SIGMA/2+(I-.5)*DS.LT.RX(1,J,K).AND.
     :              RX(1,J,K).LT.-SIGMA/2+(I+.5)*DS) THEN
                  DIST(I)=DIST(I)+1.D0
               ENDIF
            ENDDO
         ENDDO
         WRITE(10,*) -SIGMA/2+I*DS,DIST(I)/NORM
      ENDDO

      CLOSE(UNIT=10)

      RETURN
      END


      FUNCTION FACT ( N )
      IMPLICIT NONE
C     FACTORIAL FUNCTION
      INTEGER*8 FACT,N,P,I
      P=1
      DO I=1,N
         P=P*I
      ENDDO
      FACT=P
      END


      SUBROUTINE SUMUP (POTK, OVRLAP, KE, V, W)
      IMPLICIT NONE
C     ****************************************************************
C     ** CALCULATES THE TOTAL ENERGY                                **
C     **                                                            **
C     ** USAGE:                                                     **
C     **                                                            **
C     ** THE SUBROUTINE RETURNS THE TOTAL ENERGY AT THE             **
C     ** BEGINNING AND END OF THE RUN.                              **
C     ****************************************************************
      INCLUDE       'mc-mix.par'

      REAL*8        V, KE, VV, KK
      LOGICAL       OVRLAP
      CHARACTER     POTK*(*)

      REAL*8        RXII, RXIJ
      REAL*8        VIJ, WIJ, RIJSQ, W, WW

      INTEGER*8     TAU, I, J, IDIM

C     POTENTIAL ACTION

      VV      = 0.0
      WW      = 0.0

C     ** LOOP OVER ALL THE PAIRS IN THE LIQUID **

      DO TAU = 1, FTN0
         DO 100 I = 1, NP - 1
            DO 99 J = I + 1, NP
               RIJSQ = 0.d0
               DO IDIM = 1, DIM
                  RXIJ = RX(IDIM,TAU,I) - RX(IDIM,TAU,J)
C     ** MINIMUM IMAGE THE PAIR SEPARATIONS **
                  RXIJ = RXIJ -
     :                 DNINT ( RXIJ/SIGMA )*SIGMA
                  RIJSQ = RIJSQ + RXIJ * RXIJ
               ENDDO

               CALL POT (I, J, RIJSQ, VIJ, WIJ, POTK)
               VV   = VV + VIJ
               WW   = WW + WIJ
 99         CONTINUE
 100     CONTINUE
         V=VV
```

```
         W=WW
      ENDDO

C    KINETIC ACTION

      KK     = 0.0

      DO TAU = 1, FTNO
         DO I = 1, NP
            DO IDIM = 1, DIM
               RXII = RX(IDIM,TAU,I)-RX(IDIM,TAU+1,I)
               RXII = RXII -
     :              DNINT ( RXII/SIGMA )*SIGMA
               KK=KK+FTM(ISP(I))*(RXII**2.)/(2.DO*LS**2.)
            ENDDO
         ENDDO
         KE=KK
      ENDDO

      RETURN
      END


      SUBROUTINE PENERGY ( POTK, RXI, I, TAU,
     :   V, W )
      IMPLICIT NONE
C    ****************************************************************
C    ** RETURNS THE POTENTIAL ENERGY OF ATOM I WITH ALL OTHER ATOMS.  **
C    **                                                              **
C    ** PRINCIPAL VARIABLES:                                         **
C    **                                                              **
C    ** INTEGER I                THE ATOM OF INTEREST                **
C    ** INTEGER NP               THE NUMBER OF ATOMS                 **
C    ** INTEGER TAU              THE TIMESTEP                        **
C    ** REAL*8    RX, RY, RZ       THE ATOM POSITIONS                **
C    ** REAL*8    RXI,RYI,RZI      THE COORDINATES OF ATOM I         **
C    ** REAL*8    V                THE POTENTIAL ENERGY OF ATOM I    **
C    ** REAL*8    W                THE VIRIAL OF ATOM I              **
C    **                                                              **
C    ** USAGE:                                                       **
C    **                                                              **
C    ** THIS SUBROUTINE IS USED TO CALCULATE THE CHANGE OF ENERGY     **
C    ** DURING A TRIAL MOVE OF ATOM I. IT IS CALLED BEFORE AND        **
C    ** AFTER THE RANDOM DISPLACEMENT OF I.                          **
C    ****************************************************************
      INCLUDE     'mc-mix.par'

      REAL*8        RXI(MDIM), V, W
      INTEGER*8     I, J, TAU, IDIM
      CHARACTER     POTK*(*)

      REAL*8        RXIJ, RIJSQ, VIJ, WIJ

      V      = 0.0
      W      = 0.0

C    ** LOOP OVER ALL MOLECULES EXCEPT I  **

      DO 100 J = 1, NP

         IF ( I .NE. J ) THEN
            RIJSQ = 0.d0
            DO IDIM = 1, DIM
               RXIJ = RXI(IDIM) - RX(IDIM,TAU,J)

               RXIJ = RXIJ -
     :              DNINT ( RXIJ/SIGMA )*SIGMA
               RIJSQ = RIJSQ + RXIJ * RXIJ
            ENDDO

            CALL POT (I, J, RIJSQ, VIJ, WIJ, POTK)
            V   = V + VIJ
            W   = W + WIJ

 90      ENDIF

100   CONTINUE

      RETURN
      END


      SUBROUTINE PPPNERGY ( POTK, RXI, I, RXJ, J,
     :    TAU, V, W )
      IMPLICIT NONE
C    ****************************************************************
C    ** RETURNS THE POTENTIAL ENERGY OF ATOMS I AND J WITH            **
C    ** ALL OTHER ATOMS PLUS THE ONE BETWEEN I AND J                  **
C    **                                                              **
C    ** PRINCIPAL VARIABLES:                                         **
C    **                                                              **
C    ** INTEGER I, J             THE ATOMS OF INTEREST               **
C    ** INTEGER NP               THE NUMBER OF ATOMS                 **
C    ** INTEGER TAU              THE TIMESTEP                        **
C    ** REAL*8    RX, RY, RZ       THE ATOM POSITIONS                **
C    ** REAL*8    RXI,RYI,RZI      THE COORDINATES OF ATOM I         **
C    ** REAL*8    RXJ,RYJ,RZJ      THE COORDINATES OF ATOM J         **
C    ** REAL*8    V                THE POTENTIAL ENERGY OF ATOM I    **
C    ** REAL*8    W                THE VIRIAL OF ATOM I              **
C    **                                                              **
C    ** USAGE:                                                       **
C    **                                                              **
C    ** THIS SUBROUTINE IS USED TO CALCULATE THE CHANGE OF ENERGY     **
C    ** DURING A TRIAL MOVE OF ATOM I. IT IS CALLED BEFORE AND        **
C    ** AFTER THE RANDOM DISPLACEMENT OF I.                          **
C    ****************************************************************
      INCLUDE     'mc-mix.par'

      REAL*8        RXI(MDIM), RXJ(MDIM), V, W
      INTEGER*8     I, J, K, TAU, IDIM
      CHARACTER     POTK*(*)

      REAL*8        RXIJ, RIJSQ, VIJ, WIJ

      V      = 0.0
      W      = 0.0

C    ** LOOP OVER ALL MOLECULES EXCEPT I AND J **

      DO 100 K = 1, NP
         IF ( K .NE. I .AND. K .NE. J ) THEN
            RIJSQ = 0.d0
            DO IDIM = 1, DIM
               RXIJ = RXI(IDIM) - RX(IDIM,TAU,K)

               RXIJ = RXIJ -
     :              DNINT ( RXIJ/SIGMA )*SIGMA
               RIJSQ = RIJSQ + RXIJ * RXIJ
            ENDDO

            CALL POT (I, K, RIJSQ, VIJ, WIJ, POTK)
            V   = V + VIJ
            W   = W + WIJ

            IF ( I .NE. J ) THEN
               RIJSQ = 0.d0
               DO IDIM = 1, DIM
                  RXIJ = RXJ(IDIM) - RX(IDIM,TAU,K)

                  RXIJ = RXIJ -
     :                 DNINT ( RXIJ/SIGMA )*SIGMA
                  RIJSQ = RIJSQ + RXIJ * RXIJ
               ENDDO

               CALL POT (K, J, RIJSQ, VIJ, WIJ, POTK)
               V   = V + VIJ
               W   = W + WIJ

            ENDIF

         ENDIF

 100   CONTINUE

C       RETURN

      IF (I .NE. J) THEN
         RIJSQ = 0.d0
         DO IDIM = 1, DIM
            RXIJ = RXI(IDIM) - RXJ(IDIM)

            RXIJ = RXIJ -
     :            DNINT ( RXIJ/SIGMA )*SIGMA
            RIJSQ = RIJSQ + RXIJ * RXIJ
         ENDDO

         CALL POT (I, J, RIJSQ, VIJ, WIJ, POTK)
         V   = V + VIJ
         W   = W + WIJ
      ENDIF

      RETURN
      END


      SUBROUTINE KENERGY ( RXI, I, TAU, KE )
      IMPLICIT NONE
C    ****************************************************************
C    ** RETURNS THE KINETIC ENERGY OF ATOM I WITH ALL OTHER ATOMS.    **
C    **                                                              **
C    ** PRINCIPAL VARIABLES:                                         **
C    **                                                              **
C    ** INTEGER I                THE ATOM OF INTEREST                **
C    ** INTEGER NP               THE NUMBER OF ATOMS                 **
C    ** INTEGER TAU              THE TIMESTEP                        **
C    ** REAL*8    RX, RY, RZ       THE ATOM POSITIONS                **
C    ** REAL*8    RXI,RYI,RZI      THE COORDINATES OF ATOM I         **
C    ** REAL*8    KE               THE KINETIC ENERGY OF ATOM I      **
C    **                                                              **
C    ** USAGE:                                                       **
C    **                                                              **
C    ** THIS SUBROUTINE IS USED TO CALCULATE THE CHANGE OF ENERGY     **
C    ** DURING A TRIAL MOVE OF ATOM I. IT IS CALLED BEFORE AND        **
C    ** AFTER THE RANDOM DISPLACEMENT OF I.                          **
C    ****************************************************************
      INCLUDE     'mc-mix.par'

      REAL*8        RXI(MDIM), KE
      REAL*8        RXIP, RXIS, RXII
      INTEGER*8     I, TAU, IDIM

      KE = 0.d0
      DO IDIM = 1, DIM
         RXIP = RX(IDIM,TAU-1,I)
         RXIS = RX(IDIM,TAU+1,I)
         RXII = RXI(IDIM) - RXIP
         RXII = RXII - DNINT ( RXII/SIGMA )*SIGMA
         KE=KE+0.5*FTM(ISP(I))*(RXII/LS)**2.
         RXII = RXI(IDIM) - RXIS
         RXII = RXII - DNINT ( RXII/SIGMA )*SIGMA
         KE=KE+0.5*FTM(ISP(I))*(RXII/LS)**2.
      ENDDO

      RETURN
      END


      SUBROUTINE KKKNERGY ( RXI, I, TAU, KE )
      IMPLICIT NONE
C    ****************************************************************
C    ** RETURNS THE KINETIC ENERGY OF ATOM I WITH ALL OTHER ATOMS.    **
C    **                                                              **
```

```
C    ** PRINCIPAL VARIABLES:                                          **
C    **                                                               **
C    ** INTEGER I                  THE ATOM OF INTEREST               **
C    ** INTEGER NP                 THE NUMBER OF ATOMS                **
C    ** INTEGER TAU                THE TIMESTEP                       **
C    ** REAL*8    RX, RY, RZ         THE ATOM POSITIONS               **
C    ** REAL*8    RXI,RYI,RZI        THE COORDINATES OF ATOM I        **
C    ** REAL*8    KE                 THE KINETIC ENERGY OF ATOM I     **
C    **                                                               **
C    ** USAGE:                                                        **
C    **                                                               **
C    ** THIS SUBROUTINE IS USED TO CALCULATE THE CHANGE OF ENERGY     **
C    ** DURING A TRIAL MOVE OF ATOM I. IT IS CALLED BEFORE AND        **
C    ** AFTER THE RANDOM DISPLACEMENT OF I.                           **
C    ******************************************************************
        INCLUDE     'mc-mix.par'

        REAL*8        RXI(MDIM), KE
        REAL*8        RXIS, RXII
        INTEGER*8     I, TAU, IDIM

        KE = 0.d0
        DO IDIM = 1, DIM
           RXIS = RX(IDIM,TAU+1,I)
           RXII = RXI(IDIM) - RXIS
           RXII = RXII - DNINT ( RXII/SIGMA )*SIGMA
           KE=KE+0.5*FTM(ISP(I))*(RXII/LS)**2.
        ENDDO

        RETURN
        END


        SUBROUTINE CWIND ( ISPEC, WIND )
        IMPLICIT NONE
C    ******************************************************************
C    ** RETURNS THE WINDING NUMBER.                                   **
C    ******************************************************************
        INCLUDE     'mc-mix.par'

        REAL*8        WIND(MMIX,MDIM), WI, RX1, RX2
        INTEGER*8     I, J, IDIM, ISPEC

        WIND = 0.d0
        DO I = 1, NP
           IF (ISP(I).NE.ISPEC) GOTO 10
           DO IDIM = 1, DIM
              DO J = 1, FTN0 - 1
                 RX1 = RX(IDIM,J  ,I)
                 RX2 = RX(IDIM,J+1,I)
                 WI  = RX2-RX1-DNINT((RX2-RX1)/SIGMA)*SIGMA
                 WIND(ISPEC,IDIM) = WIND(ISPEC,IDIM) + WI
              ENDDO
           ENDDO
 10        CONTINUE
        ENDDO

        RETURN
        END


        REAL*8 FUNCTION RANF ( DUMMY )
        IMPLICIT NONE
C    ******************************************************************
C    ** RETURNS A UNIFORM RANDOM VARIATE IN THE RANGE 0 TO 1.         **
C    **                                                               **
C    **                 ***************                               **
C    **                 **  WARNING  **                               **
C    **                 ***************                               **
C    **                                                               **
C    ** GOOD RANDOM NUMBER GENERATORS ARE MACHINE SPECIFIC.           **
C    ** PLEASE USE THE ONE RECOMMENDED FOR YOUR MACHINE.              **
C    **                                                               **
C    ** RAND(FLAG) returns a pseudo-random number from a uniform      **
C    ** distribution between 0 and 1. If FLAG is 0, the next number   **
C    ** in the current sequence is returned; if FLAG is 1, the        **
C    ** generator is restarted by CALL SRAND(0); if FLAG has any      **
C    ** other value, it is used as a new seed with SRAND.             **
C    ******************************************************************
        INTEGER*8   L, C, M
        PARAMETER ( L = 1029, C = 221591, M = 1048576 )
        INTEGER*8   SEED
        SAVE        SEED
        DATA        SEED / 0 /
        REAL*8      DUMMY

C        SEED = MOD ( SEED * L + C, M )
C        RANF = DBLE( SEED ) / M

        RANF = RAND ( )

        RETURN
        END


        SUBROUTINE READCN ( CNFILE )
        IMPLICIT NONE
C    ******************************************************************
C    ** SUBROUTINE TO READ IN THE CONFIGURATION FROM UNIT 10          **
C    ******************************************************************
        INCLUDE     'mc-mix.par'

        CHARACTER   CNFILE*(*), HASH*1, MYFMT*77

        INTEGER*8   CNUNIT, I, J, NNP, NI, IDIM
        PARAMETER ( CNUNIT = 10 )

C   ******************************************************************

        OPEN ( UNIT = CNUNIT, FILE = CNFILE, STATUS = 'OLD' )
        WRITE(MYFMT,'(A,I10,A)') '(I3,3X,',DIM,'(F12.6,3X),I3)'

        READ ( CNUNIT,* ) HASH, FTN0, NNP
        IF ( NNP .NE. NP ) STOP 'N ERROR IN READCN'

        DO 100 I = 1, NNP
           READ ( CNUNIT,* ) HASH, NEXT(I)
           NI = NEXT(I)
           DO 90 J = 1, FTN0+1
              READ ( CNUNIT, MYFMT) LEXT(I), (RX(IDIM,J,I),IDIM=1,DIM),
     :                              ISP(I)
 90        ENDDO
           READ ( CNUNIT,13 )
           READ ( CNUNIT,13 )
           DO IDIM = 1, DIM
              RX(IDIM,0,NI) = RX(IDIM,FTN0,I)
           ENDDO
 100    ENDDO
 13     FORMAT(A2)

        CLOSE ( UNIT = CNUNIT )

        RETURN
        END


        SUBROUTINE WRITCN ( CNFILE )
        IMPLICIT NONE
C    ******************************************************************
C    ** SUBROUTINE TO WRITE OUT THE CONFIGURATION TO UNIT 10          **
C    ******************************************************************
        INCLUDE     'mc-mix.par'

        CHARACTER   CNFILE*(*), MYFMT*77

        INTEGER*8   CNUNIT, I, J, IDIM
        PARAMETER ( CNUNIT = 10 )

C   ******************************************************************

        OPEN ( UNIT = CNUNIT, FILE = CNFILE, STATUS = 'UNKNOWN' )
        WRITE(MYFMT,'(A,I10,A)') '(I3,3X,',DIM,'(F12.6,3X),I3)'

        WRITE(*,*) 'output to file ------------------------------------
     :---------------'
        CALL FLUSH ( CNUNIT )
        WRITE ( CNUNIT,* ) '#',FTN0,NP

        DO I = 1, NP
           WRITE ( CNUNIT,* ) '#', NEXT(I)
           DO J = 1, FTN0+1
              WRITE ( CNUNIT, MYFMT ) LEXT(I),
     :      (RX(IDIM,J,I)-DNINT(RX(IDIM,J,I)/SIGMA)*SIGMA,IDIM=1,DIM),
     :      ISP(I)
C               WRITE ( CNUNIT, MYFMT ) LEXT(I),
C     :       (RX(IDIM,J,I),IDIM=1,DIM), ISP(I)
           ENDDO
           WRITE ( CNUNIT,13 ) ''
           WRITE ( CNUNIT,13 ) ''
        ENDDO
 13     FORMAT(A2)

        CLOSE ( UNIT = CNUNIT )

        RETURN
        END


        SUBROUTINE INITCN ( CNFILE )
        IMPLICIT NONE
C    ******************************************************************
C    ** SUBROUTINE TO INITIALIZE THE CONFIGURATION TO UNIT 10         **
C    ******************************************************************
        INCLUDE     'mc-mix.par'

        REAL*8      RANF
        CHARACTER   CNFILE*(*), MYFMT*77

        INTEGER*8   CNUNIT, I, J, IDIM, I1, IP, NNP
        REAL*8      RXI(DIM), RXII(DIM, NP), RXIJ, RIJSQ
        PARAMETER ( CNUNIT = 10 )

C   ******************************************************************

        OPEN ( UNIT = CNUNIT, FILE = CNFILE, STATUS = 'UNKNOWN' )
        WRITE(MYFMT,'(A,I10,A)') '(I3,3X,',DIM,'(F12.6,3X),I3)'

        WRITE ( CNUNIT,* ) '#',FTN0,NP
        RXII = 0.d0

        IP = 1
        NNP = NPM(IP)
        DO I = 1, NP
           ISP(I)=IP
           IF (I.GE.NNP) THEN
              IP = IP + 1
              NNP = NNP + NPM(IP)
           ENDIF
        ENDDO

        DO I = 1, NP
           WRITE ( CNUNIT,* ) '#', NEXT(I)
 10        DO IDIM = 1, DIM
              RXI(IDIM) = SIGMA*(RANF(0.d0)-.5)
           ENDDO
           DO I1 = 1, NP
              RIJSQ = 0.d0
              DO IDIM = 1, DIM
                 RXIJ = RXI(IDIM)-RXII(IDIM,I1)
                 RXIJ = RXIJ  -
     :                DNINT ( RXIJ/SIGMA )*SIGMA
                 RIJSQ = RIJSQ + RXIJ * RXIJ
```

```
            ENDDO
            IF (RIJSQ.LE.EPSA(ISP(I),ISP(I1))**2) THEN
               GOTO 10
            ENDIF
         ENDDO
         DO IDIM = 1, DIM
            RXII(IDIM,I) = RXI(IDIM)
         ENDDO
         DO J = 1, FTNO+1
            WRITE ( CNUNIT, MYFMT ) I, (RXI(IDIM),IDIM=1,DIM), ISP(I)
            DO IDIM = 1, DIM
               RX(IDIM,J,I) = RXI(IDIM)
            ENDDO
         ENDDO
         WRITE ( CNUNIT,13 ) ''
         WRITE ( CNUNIT,13 ) ''
      ENDDO
 13   FORMAT(A2)

      CLOSE ( UNIT = CNUNIT )

      RETURN
      END


      SUBROUTINE POT (I, J, RIJSQ, VIJ, WIJ, POTK)
      IMPLICIT NONE
C    ****************************************************************
C    ** SUBROUTINE FOR THE PAIR POTENTIAL                          **
C    ****************************************************************
      INCLUDE     'mc-mix.par'

      INTEGER*8 I, J, K
      CHARACTER POTK*(*)
      REAL*8    RIJ, RIJSQ, RMIN, RMSQ, RCSQ, SR2, SR6, VIJ, WIJ
      REAL*8    COU3, QHS3, F
      PARAMETER ( COU3 = 4.760154727959107013287634700 57d0 )
      PARAMETER ( QHS3 = 24.d0 )

C       PI = ACOS(-1.d0)

      VIJ = 0.d0
      WIJ = 0.d0
      IF (DIM .GT. 3) THEN
         WRITE(*,*) "POTENITAIL CANNOT HAVE DIM > 3 !!!"
         STOP
      ENDIF
      IF (POTK .EQ. 'FREE') RETURN

      IF (POTK .EQ. 'LJ') THEN
CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC
C     W. L. McMillan, Phys. Rev. A 138, 442 (1965)     C
CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC
C          RMIN = TINY(PI)
         RMIN = 0.d0
         RIJSQ = RIJSQ / (SIG(I,J) * SIG(I,J))
         RMSQ = RMIN*RMIN
         RCSQ = RCUT*RCUT
         IF (RIJSQ .LT. RMSQ) THEN
            RIJSQ = RMSQ
            SR2 = 1.d0 / RIJSQ
            SR6 = SR2 * SR2 * SR2
            VIJ = SR6 * ( SR6 - 1.d0 )
            WIJ = SR6 * ( SR6 - 5.d-1 )
            VIJ = 4.d0 * EPS * VIJ
            WIJ = 48.d0 * EPS * WIJ / 3.d0
         ELSEIF (RIJSQ .GT. RMSQ .AND. RIJSQ .LT. RCSQ) THEN
            SR2 = 1.d0 / RIJSQ
            SR6 = SR2 * SR2 * SR2
            VIJ = SR6 * ( SR6 - 1.d0 )
            WIJ = SR6 * ( SR6 - 5.d-1 )
            VIJ = 4.d0 * EPS * VIJ
            WIJ = 48.d0 * EPS * WIJ / 3.d0
         ENDIF
      ELSEIF (POTK .EQ. 'BUMP') THEN
         RIJ = SQRT( RIJSQ )
         IF (RIJ .LE. SIG(I,J)) THEN
            VIJ = EPSR
         ENDIF
      ELSEIF (POTK .EQ. 'PSW') THEN
         RIJ = SQRT( RIJSQ )
         IF (RIJ .LE. SIG(I,J)) THEN
            VIJ = EPSR
         ELSEIF (RIJ .LE. SIG(I,J)+RCUT .AND. RIJ .GT. SIG(I,J)) THEN
            VIJ = -EPSA(I,J)
         ENDIF
      ELSEIF (POTK .EQ. 'COULOMBHS') THEN
c          F=0.
c          DO K=1,NMIX
c             F = F+SIG(1,K)*NPM(K)/(NPM(K)-1)
c          ENDDO
         RIJ = SQRT( RIJSQ )
         IF (RIJ .GT. EPSA(ISP(I),ISP(J))) THEN
            IF (DIM .EQ. 3) THEN
c               VIJ = -F*COU3*RCUT/SIGMA/2.
c               VIJ = VIJ + RCUT*/RIJ
               VIJ = SIG(ISP(I),ISP(J))*RCUT/RIJ
            ELSEIF (DIM .EQ. 2) THEN
c               VIJ = -F*(6.-PI+LOG(4.)-4.*LOG(SIGMA/RCUT))/4.
c               VIJ = VIJ - LOG(RIJ/RCUT)
               VIJ = -SIG(ISP(I),ISP(J))*LOG(RIJ/RCUT)
            ELSEIF (DIM .EQ. 1) THEN
c               VIJ = F*SIGMA/RCUT/4.
c               VIJ = VIJ - RIJ/RCUT
               VIJ = -SIG(ISP(I),ISP(J))*RIJ/RCUT
            ENDIF
         ELSE
            VIJ = 10.**10.
         ENDIF
      ELSEIF (POTK .EQ. 'COULOMB') THEN
c          F=0.
c          DO K=1,NMIX
c             F = F+SIG(1,K)*NPM(K)/(NPM(K)-1)
c          ENDDO
         RIJ = SQRT( RIJSQ )
         IF (RIJ .GT. EPSA(ISP(I),ISP(J))) THEN
            IF (DIM .EQ. 3) THEN
c               VIJ = -F*COU3*RCUT/SIGMA/2.
c               VIJ = VIJ + RCUT*/RIJ
               VIJ = SIG(ISP(I),ISP(J))*RCUT/RIJ
            ELSEIF (DIM .EQ. 2) THEN
c               VIJ = -F*(6.-PI+LOG(4.)-4.*LOG(SIGMA/RCUT))/4.
c               VIJ = VIJ - LOG(RIJ/RCUT)
               VIJ = -SIG(ISP(I),ISP(J))*LOG(RIJ/RCUT)
            ELSEIF (DIM .EQ. 1) THEN
c               VIJ = F*SIGMA/RCUT/4.
c               VIJ = VIJ - RIJ/RCUT
               VIJ = -SIG(ISP(I),ISP(J))*RIJ/RCUT
            ENDIF
         ELSE
            RIJ = EPSA(ISP(I),ISP(J))
            IF (DIM .EQ. 3) THEN
c               VIJ = -F*COU3*RCUT/SIGMA/2.
c               VIJ = VIJ + RCUT*/RIJ
               VIJ = SIG(ISP(I),ISP(J))*RCUT/RIJ
            ELSEIF (DIM .EQ. 2) THEN
c               VIJ = -F*(6.-PI+LOG(4.)-4.*LOG(SIGMA/RCUT))/4.
c               VIJ = VIJ - LOG(RIJ/RCUT)
               VIJ = -SIG(ISP(I),ISP(J))*LOG(RIJ/RCUT)
            ELSEIF (DIM .EQ. 1) THEN
c               VIJ = F*SIGMA/RCUT/4.
c               VIJ = VIJ - RIJ/RCUT
               VIJ = -SIG(ISP(I),ISP(J))*RIJ/RCUT
            ENDIF
         ENDIF
      ELSEIF (POTK .EQ. 'QHS') THEN
c          F=0.
c          DO K=1,NMIX
c             F = F+SIG(1,K)*NPM(K)/(NPM(K)-1)
c          ENDDO
         IF (DIM .EQ. 3) THEN
            IF (RIJSQ .LE. SIG(I,J)*SIG(I,J)) THEN
               VIJ = 10.**10.
            ELSE
c               VIJ = -F*QHS3*(SIGMA-SIG(I,J))/SIGMA**3/2./FTM(I)
            VIJ = VIJ + (2.*RIJSQ+SIG(I,J)**2.)/
     :                  (RIJSQ-SIG(I,J)**2.)**2./2./FTM(I)
            WIJ = -2.*RIJSQ*(RIJSQ+2.*SIG(I,J)**2.)/
     :                     (RIJSQ-SIG(I,J)**2.)**3./FTM(I)
            ENDIF
         ELSE
            WRITE(*,*) 'QHS in dimension different from 3'
            STOP
         ENDIF
      ELSEIF (POTK .EQ. 'HARMONIC') THEN
         VIJ = 0.5*RIJSQ/RCUT**2.
      ELSE
         WRITE(*,*) 'NO PAIR POTENTIAL AVAILABLE!'
         STOP
      ENDIF

      RETURN
      END


      SUBROUTINE GOFR ( C, DENS, MAXBIN, HIST11, HIST12, HIST22 )
      IMPLICIT NONE
C    ****************************************************************
C    ** CALCULATES THE RADIAL DISTRIBUTION FUNCTION HISTOGRAM.     **
C    ****************************************************************
      INCLUDE     'mc-mix.par'

      REAL*8        RXIJ, RIJSQ, RIJ, DENS
      INTEGER*8     C, IDIM, I, J, BIN, MAXBIN
      INTEGER*8     HIST11(MAXBIN), HIST12(MAXBIN), HIST22(MAXBIN)

      DO I = 1, NP - 1
         DO J = I + 1, NP
            RIJSQ = 0.
            DO IDIM = 1, DIM
               RXIJ  = RX(IDIM,C,I) - RX(IDIM,C,J)
               RXIJ  = RXIJ - ANINT ( RXIJ/SIGMA ) * SIGMA
               RIJSQ = RIJSQ + RXIJ * RXIJ
            ENDDO
C     CALCULATE HISTOGRAM FOR g(r)
            RIJ = SQRT ( RIJSQ )
            BIN = INT ( RIJ/DELR ) + 1

            IF (BIN .LE. MAXBIN) THEN
               IF (ISP(I) .EQ. 1 .AND. ISP(J) .EQ. 1) THEN
                  HIST11(BIN) = HIST11(BIN) + 2
               ELSEIF(ISP(I) .EQ. 2 .AND. ISP(J) .EQ. 2) THEN
                  HIST22(BIN) = HIST22(BIN) + 2
               ELSEIF(ISP(I) .NE. ISP(J)) THEN
                  HIST12(BIN) = HIST12(BIN) + 1
               ENDIF
            ENDIF

         ENDDO
      ENDDO

      RETURN
      END


CCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCCC
c     determines the sign of the free spinless Fermi density matrix
c     dim  = spatial dimensions
c     nf   = number of fermions
c     beta = inverse temperature
c     itau0= initial time slice
c     itau1= final   time slice
```

```
c     sign = sign of the free fermion density matrix
cccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccccc
      subroutine ifdm(ispec,itau0,itau1,sign)
      implicit none
      include 'mc-mix.par'
      integer*8 ispec,itau0,itau1,i,j,ii,jj,idim,npp
      real*8 sum,DMGT,epsilon,det,sign
      real*8 aaa(npm(ispec),npm(ispec)),beta,lambda,dx

      epsilon=1.d-6
      beta=ls*abs(itau1-itau0)
      lambda=1./2./ftm(ispec)

      if (itau0.eq.itau1) then
         sign=1.
         return
      endif

      npp=0
      do i=1,ispec-1
         npp=npp+npm(isp(i))
      enddo

c     set up the free particle density matrix
      do i=1,npm(ispec)
         do j=1,npm(ispec)
            ii=i+npp
            jj=j+npp
            sum=0.d0
            do idim=1,dim
               dx=rx(idim,itau0,jj)-rx(idim,itau1,ii)
               dx=dx-dnint(dx/sigma)*sigma
               sum=sum+dx**2.
            enddo
            sum=sum/(4*beta*lambda)
            aaa(i,j)=exp(-sum)
            aaa(i,j)=aaa(i,j)/(4*pi*beta*lambda)**(dim/2.)
         enddo
      enddo

      det=DMGT(epsilon,npm(ispec),aaa)

c     sign of the determinant
      sign=det

      return
      end


!The subroutine TSRGT applies to input real square matrix A(n,n) the upper
!triangularization algorithm of Gauss method with full pivoting and keeps
!trace of successive transformations done in integer vectors KP and LP.
!-----------------------------------------------------------------------
!  Input parameters:
!  eps        precision (real*8)
!  n          size of A matrix (integer)
!  A          pointer to input real square matrix (real*8)
!  Output parameters:
!  it         flag=1 if A matrix ok, =0 if A matrix is singular (integer)
!  C          pointer to table storing main diagonal elements and supra-
!             diagonal elements of upper triangular matrix and the multi-
!             plying coefficients used during triangularization process
!  KP         table storing informations concerning the column exchanges
!             during process (integer)
!  LP         table storing informations concerning the line exchanges
!             during process (integer)
!----------------------------------------------------------------------
!The table C is first initialized to A matrix, then receives at each step k
!of the triangularization process, usefull elements of A matrix at step k for
!k=1,2,...n.
!The variables po(real*8), lo and ko(integer) store respectively pivot at step k,
!its line number and its column number.
!----------------------------------------------------------------------
      Subroutine TSRGT(eps, n, A, it, C, Kp, Lp)
      real*8 eps
      integer*8 n,it
      real*8 A(n,n), C(n,n)
      integer*8 Kp(n),Lp(n)
      real*8  po,t0
      C=A
      it=1
      k=1
      do while (it==1.and.k<n)
         po=C(k,k)
         lo=k
         ko=k
         do i=k, n
            do j=k, n
               if (dabs(C(i,j))>dabs(po)) then
                  po=C(i,j)
                  lo=i
                  ko=j
               end if
            end do
         end do
         Lp(k)=lo
         Kp(k)=ko
         if (dabs(po)<eps) then
            it=0
         else
            if (lo.ne.k) then
               do j=k, n
                  t0=C(k,j)
                  C(k,j)=C(lo,j)
                  C(lo,j)=t0
               end do
            end if
            if (ko.ne.k) then
               do i=1, n
                  t0=C(i,k)
                  C(i,k)=C(i,ko)
                  C(i,ko)=t0
               end do
            end if
            do i=k+1, n
               C(i,k)=C(i,k)/po
               do j=k+1, n
                  C(i,j)=C(i,j)-C(i,k)*C(k,j)
               end do
            end do
            k=k+1
         end if
      end do
      if (it==1.and.dabs(C(n,n))<eps)  it=0
      return
      End                       !TSRGT


!The function DMGT returns the determinant of a real square matrix
!A(n,n) by Gauss method with full pivoting.
!-----------------------------------------------------------------------
!  Input parameters:
!  eps        precision (real*8)
!  n          size of A matrix (integer)
!  A          pointer to input real square matrix
!  Output parameters:
!  None
!----------------------------------------------------------------------
!The procedure TSRGT is used to reduce A matrix to an upper triangular matrix.
!Output variables are it(integer), C(n,n), Kp(n) and Lp(n).
!If it=0, matrix A is singular, if it=1, matrix A is regular. Table C contains
!at location i,j (j>=i) the corresponding element of the upper triangular matrix.
!Tables Lp and Kp contain informations relative to exchanges of line or column
!that occured during the process. For instance, the element number k of Lp is
!an integer <> k if an exchange of line has been made at step k (k=1,2,...,n).
!The number of exchanges of lines and columns is stored in integer L. the
!determinant of A matrix is stored in d0 (real*8).
!----------------------------------------------------------------------
      real*8 Function DMGT(eps, n, A)
      integer*8 n, it, Kp(n), Lp(n)
      real*8 eps, A(n,n), C(n,n)
      real*8 d0

      call TSRGT(eps,n,A,it,C,Kp,Lp) !call triangularization subroutine
      if (it==0) then
         d0=0.d0                !matrix singular, det=0
      else                      !matrix regular, det<>0
         d0=1.d0
         do k=1, n
   d0=d0*C(k,k)
       end do
       l=0
       do k=1, n-1
          if (Lp(k).ne.k)  l=l+1
          if (Kp(k).ne.k)  l=l+1
       end do
      if (MOD(l,2).ne.0) d0=-d0 !l is odd
      end if
      DMGT=d0                   !return determinant
      return
      End


        SUBROUTINE EQUIV(A,B)
        IMPLICIT NONE
C     ****************************************************************
C     ** A = B WITH A, B TWO PATHS                                  **
C     ****************************************************************
        INCLUDE     'mc-mix.par'

        INTEGER*8 I,J,K
        REAL*8    A(MDIM,0:N,MNP),B(MDIM,0:N,MNP)

        DO I=1,DIM
           DO J=0,FTN0+1
              DO K=1,NP
                 A(I,J,K)=B(I,J,K)
              ENDDO
           ENDDO
        ENDDO

        RETURN
        END

*******************************************************************************
*** mc-mix.par ****************************************************************
*******************************************************************************

      REAL*8        PI
      REAL*8        DELR
         INTEGER*8     MDIM, N, MNP, MBIN, MMIX

         PARAMETER ( PI = 3.14159265358979323846264338328d0 )

      PARAMETER ( MDIM = 10   )   ! maximum number of dimensions
       PARAMETER ( MNP  = 1000 )   ! maximum number of particles
      PARAMETER ( N    = 3000 )   ! maximum number of time slices
      PARAMETER ( MBIN = 100000 ) ! maximum dimension of g(r) histogram
      PARAMETER ( MMIX = 2 )      ! maximum number of components
         PARAMETER ( DELR = 0.01d0 ) ! grid spacing for g(r)


      INTEGER*8     DIM, NMIX, ISP(MNP)
         REAL*8        RX(MDIM,0:N,MNP)
         REAL*8        FTM(MMIX), LS, SIGMA
         REAL*8        SIG(MMIX,MMIX),EPSA(MMIX,MMIX),EPSR,EPS,RCUT
         INTEGER*8     FTN0, NP, NPM(MMIX)
      INTEGER*8     NEXT(MNP), LEXT(MNP), MEXT(N,MNP), MBM

         ! path
         COMMON / BLOCK1 / RX, ISP, DIM, NMIX
         ! pair-potential parameters
```

```
      COMMON / BLOCK2 / SIG, EPSR, EPSA, EPS, RCUT
      ! mass, timestep, box edge, # timeslices, # particles
      COMMON / BLOCK3 / FTM, LS, SIGMA, FTN0, NP, NPM
      ! permutations
      COMMON / BLOCK4 / NEXT, LEXT, MEXT, MBM

*****************************************************************************************
*** data-mix.in *************************************************************************
*****************************************************************************************

0 spatial dimensions (1,2,3,...<= MDIM) nixture dimension (1,2,3,...<= MMIX)
3 2
1 seed of the random sequence                                          RAND
3
2 if bose                                                               (T/F)
T
3 number of particles NPM(I)                    (NP=NPM(1)+NPM(2)+...<= MNP)
10
10
4 the potential                                              SUBROUTINE POT
COULOMB
5 # time slices = 1/temperature/timestep                                (< N)
20
6 masses FTM(1), FTM(2), ...                                (hbar = kb = 1)
1.d0
1836.15d0
7 # of cycles                                                        (nstep)
500000000000000000
8 # of steps between output lines                                    (iprint)
100
9 # of steps between configuration saves                              (isave)
100
10 # of steps for equilibration                                       (iequi)
100
11 # of steps for acceptance ratios                                  (iratio)
100
12 # of steps for rdf calculation                                    (icalcg < #5)
10
13 configuration file name
conf.xyz
14 0 initialization, 1 restart, 2 restart rdf
0
15 density                                                   THERMODYNAMICS
.16d0
16 temperature                                               THERMODYNAMICS
.005d0
17 maximum displacement/box edge
.01d0
18 maximum # of bridge timeslices                              (> 1; <= #5)
20
19 potential cutoff distance (LJ)                            SUBROUTINE POT
1.d0
20 sig(i,j)                                   (LJ 2.566)     SUBROUTINE POT
1.d0
-1.d0
-1.d0
1.d0
21 epsr                                       (LJ INIT)      SUBROUTINE POT
1.d0
22 epsa(i,j)                                                 SUBROUTINE POT
0.d0
.5d0
.5d0
1.d0
23 eps                                        (LJ 10.22)     SUBROUTINE POT
1.d1
24 if only displace                                                   (T/F)
F
25 if only bridge                                                     (T/F)
F
```


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